How a Predictive State Observer Can Self-Adapt Its Sensory Prediction-Error Correction Gain Closed-Loop Evidence from a Muscle-Driven Reaching Task

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***Abstract*—We ask whether a forward-model-based predictive state observer can adapt its sensory prediction-error correction gain (hereafter, *correction gain*) from signals the agent itself generates online — innovation history and per-episode task outcome — rather than from the per-condition oracle labels that supervised gain predictors require at training time (Appendix C). A residual MLP forward model combines with a fixed-lag predictive state observer over a 34-muscle MyoSuite reaching arm; the correction gain $K \in [0, 1]$ enters in an innovation-style update. We propose a two-layer adaptation rule. *Within-trial*, a per-channel exponentially-weighted variance of squared innovation gives a reliability estimate $r_f(t)$, and a logistic $K_f(t) = \sigma(\beta_{0,f} + \beta_{1,f} \log r_f(t))$ maps it to a field-wise correction gain on the qpos, qvel, act, tip_pos, and reach_err sensory channels. *Across-trial*, Simultaneous Perturbation Stochastic Approximation (SPSA) updates the meta-parameters β from the per-episode reaching outcome. Plugging this observer into a joint-space PD controller reveals four findings. *First*, default within-trial reliability adaptation produces strongly field-heterogeneous intermediate-to-high gains rather than the task-optimal $K=0$; under a multi-step ($H=8$) supervised forward model whose closed-loop optimum is $K=0$, this rule is structurally misaligned and trails $K=0$ by 23–29 cm. *Second*, across-trial SPSA on a single delay-18 cell *closes most of the default-reliability gap on that cell*: deploying the trained β on independent episodes improves min-tip from 0.77 m (default reliability) to 0.56 m against the $K=0$ baseline of 0.50 m, recovering about 17 of the 23 cm gap; the smoothed training trajectory itself briefly reaches the $K=0$ level (≈ 0.49 m last-20-iter mean). Field-wise K specialises: qpos takes both the most-negative intercept, $\beta_{0,\mathrm{qpos}} \approx -1.5$, and the steepest slope, $\beta_{1,\mathrm{qpos}} \approx +0.9$. The third and fourth findings expose the *boundary conditions* of this single-β recipe. *Third*, multi-cell SPSA over the full grid improves over default reliability by only 3–5 cm at delay-18 cells and not at all at delay-0 cells: a single global β cannot simultaneously fit cells whose reliability geometries differ, which we read as evidence that the adaptation parameterisation, not the SPSA loop, is underpowered. *Fourth*, transferring the ReachFixed-trained β to the variable-target ReachRandom variant *hurts*: where the closed-loop oracle $K^\star_{\mathrm{cl}}$ is multimodal across cells, the transferred β trails $K=0$ at every delay-18 cell by $\sim$4–5 cm, while the same default reliability rule is competitive with $K=0$ (slightly better at 2 of 3 delay-18 cells by 1–3 cm). The correction-gain rule is therefore task-dependent: a β tuned on a uniform-$K^\star_{\mathrm{cl}}$ task cannot be reused on a multimodal-$K^\star_{\mathrm{cl}}$ task, and the right model class is context-conditioned β rather than a single global one. Offline-oracle-supervised gain predictors (Appendix C) serve as an upper-bound diagnostic against which agent-available adaptation can be measured.**

***Index Terms*—Forward model, predictive state observer, sensory prediction-error correction, reliability-adaptive correction gain, outcome-driven adaptation, SPSA, multi-step rollout supervision, closed-loop control, MyoSuite, muscle-driven reaching.**

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I. Introduction

State estimation in muscle-driven manipulation faces three coupled challenges: sensorimotor delay, observation noise, and signal-dependent motor noise. The internal-model view of biological motor control answers these with a *forward-model-based predictive state observer*: an internal forward model uses efference copy to predict the next state, and the observer corrects that prediction by a scalar weight $K \in [0, 1]$ on the sensory prediction error [1], [2]. $K=0$ trusts the forward prediction (efference copy); $K=1$ trusts the sensor; intermediate K implements a sensory-prediction-error correction. The innovation-update form is structurally familiar from Kalman filtering, where the gain is derived from propagated covariance estimates [3]–[5]; we do *not* propagate covariances and do not claim to implement a full Kalman filter (Sec. III-C). Instead we treat K as a scalar (or field-wise vector) that controls how strongly sensory prediction error corrects the current state estimate, and ask: *can the agent learn to set this gain from signals it generates online — innovation history and per-episode task outcome — rather than from the per-condition K-sweep oracle labels that supervised gain predictors use at training time (Appendix C)?*

This question is biologically motivated. The nervous system does not have access to a swept oracle $K^\star$ across noise and delay regimes; it has access to its own prediction errors and, after moving, to whether the limb arrived where intended. We therefore propose a *two-layer reliability-adaptive observer*. *Within-trial*, a per-channel exponentially-weighted variance of squared innovation gives a running reliability estimate $r_f(t)$, and a logistic $K_f(t) = \sigma(\beta_{0,f} + \beta_{1,f} \log r_f(t))$ maps it to a field-wise correction gain on the qpos, qvel, act, tip_pos, and reach_err sensory channels. *Across-trial*, Simultaneous Perturbation Stochastic Approximation (SPSA) [6] updates the meta-parameters $\beta = \{\beta_{0,f}, \beta_{1,f}\}$ from the per-episode reaching outcome. Neither layer accesses the optimal K, the true state, or a swept oracle; both use only signals the agent generates during reaching.

We instantiate this observer on a 34-muscle MyoSuite reaching arm [7] running in the MuJoCo simulator [8] with a residual-MLP forward model trained under H-step rollout supervision ($H{=}8$) and a fixed-lag predictive state observer that handles configurable observation delay ($d \in \{0, 18\}$ steps) and configurable per-field Gaussian noise. A separate joint-space PD controller with a one-shot inverse-kinematics target then *deploys* the observer in closed loop. As a diagnostic upper bound, we also retain an offline-oracle-supervised correction-gain predictor (Appendix C), which uses a per-condition closed-loop K-sweep to define $K^\star_{\text{cl}}$ and is the strongest non-agent-available baseline available in this setting.

Our contributions are summarised in four claims.

- **C1.** *Default within-trial reliability adaptation is sensible but task-misaligned.* With $\beta_{0,f} = 0$ and $\beta_{1,f} = 0.5$ (a deliberately neutral prior), the within-trial rule produces strongly field-heterogeneous intermediate-to-high gains rather than the task-optimal $K{=}0$; qvel is the only consistently suppressed field, while other channels often remain near sensor-trusting values (Table I). Under the $H{=}8$ forward model on this task, however, the closed-loop oracle $K^\star_{\text{cl}}$ is $K{=}0$ at every cell, so default reliability adaptation trails $K{=}0$ by 23–29 cm in $\mathrm{min\text{-}tip}$ error (Fig. 2). Innovation-based reliability alone — without an outcome signal — cannot recover the right level on this task.
- **C2.** *Single-cell across-trial SPSA closes the gap.* Running SPSA on the per-episode $\mathrm{min\text{-}tip}$ outcome on the single ($\sigma{=}\mathrm{none}$, $d{=}18$) cell for 100 iterations (12 samples/side, paired perturbations) drives β to a regime in which the field-wise correction gains specialize: qpos becomes both the most-suppressed ($\beta_{0,\mathrm{qpos}} \approx -1.5$) and the most-reactive ($\beta_{1,\mathrm{qpos}} \approx 0.9$) field, driving K_{qpos} deepest under high innovation variance. On the training cell the smoothed training-time outcome falls to ≈ 0.49 m (last-20-iter mean), within 1 cm of the $K{=}0$ baseline (0.50 m on the same cell) and well below default reliability (0.77 m). Outcome-driven β adaptation alone, without per-condition oracle labels, can recover $K^\star_{\text{cl}}$-equivalent performance.
- **C3.** *Multi-cell SPSA is bottlenecked by cell heterogeneity.* Repeating the same SPSA on the full 6-cell grid (uniform random cell sampling per iteration) yields a balanced single β with $K_{\mathrm{mean}} \approx 0.51$ at delay-18 cells and ≈ 0.63 at delay-0 cells. The resulting outcome improvement at delay-18 cells is only 3–5 cm, and there is no improvement at delay-0 cells. A single global β cannot simultaneously suppress K where useful and elevate it elsewhere unless $\beta_{0,f}$ is pushed deeply negative; the structural cost of one-β-fits-all surfaces as residual outcome error (Fig. 4).
- **C4.** *Task transfer reveals the rule is task-dependent.* On the variable-target `ReachRandom` variant we find that $K^\star_{\text{cl}}$ is multimodal across cells ($\{0, 0.25, 1\}$, Appendix F); a single K no longer dominates. Transferring the ReachFixed-trained β to ReachRandom *hurts* (pushes K too low globally and worsens $\mathrm{min\text{-}tip}$ by 4–5 cm at delay-18), while the *default* within-trial reliability rule alone — with no SPSA training of β — matches or beats $K{=}0$ at 2 of 3 delay-18 cells (by 1–3 cm). The right correction-gain rule is therefore task-dependent: outcome-driven β adaptation matches when $K^\star_{\text{cl}}$ is uniform; default reliability is competitive when $K^\star_{\text{cl}}$ is heterogeneous.

Together, C1–C4 say that the closed-loop correction gain can be recovered from agent-available signals alone, but only when the within-trial reliability rule, the across-trial outcome objective, and the task's reliability–outcome geometry are in agreement. They also expose a structural limitation of any single-β adaptation strategy on tasks where $K^\star_{\text{cl}}$ varies across noise–delay–target cells. Appendix C reports an offline-oracle-supervised correction-gain predictor as the strongest non-agent-available baseline; that predictor is a useful upper-bound diagnostic against which our agent-available adaptation can be measured, but as a model of how a biological agent could set its correction gain it requires access the agent does not have.

The rest of the paper is organised as follows. Section II positions our work against prior literature on adaptive correction gains and Kalman-style updates, forward models in motor control, learned dynamics, outcome-driven adaptation, and muscle-driven simulation. Section III describes the task, the forward model, the fixed-lag predictive state observer, the reliability-adaptive correction-gain rule, the across-trial SPSA loop, the controller, and the evaluation pipeline. Section IV reports the four claims with the corresponding figures. Section V discusses the trade-offs and the implications of agent-available correction-gain adaptation for biological motor control models.

II. Related Work

Adaptive correction gains and Kalman-style updates. The Kalman filter [3] and its smoothing counterpart [9] are the textbook framework for combining a state-space model with noisy observations under known noise covariances. When the covariances are unknown, the classical *adaptive Kalman filter* estimates them online from the innovation sequence [4]; the steady-state gain K then drops out of the covariance estimate. We adopt the same innovation-style update form (residual MLP for dynamics, fixed-lag buffered correction for delays) but *do not* propagate covariances and do not compute an optimal Kalman gain; the state-space formulation is standard textbook material [5] and we treat the result as a predictive state observer with a scalar correction gain (Sec. III-C). The novelty of our setting is therefore not in the filter equations but in the *choice* of K: instead of deriving K from a noise-covariance estimate, we train a small condition-level predictor that maps (controller, σ, delay) to a scalar K supervised against a swept oracle. The contribution we report is what happens when that oracle is open-loop versus closed-loop, a distinction largely absent from the adaptive-Kalman literature and not natural inside the covariance-propagation framing.

Forward / internal models in motor control. The view that the central nervous system maintains an internal forward model of the limb to predict the sensory consequences of motor commands traces back to Bernstein's degrees-of-freedom analysis [1] and was given an explicit computational framing by Wolpert and Ghahramani [2], who identified forward and inverse models as complementary processes underlying sensorimotor estimation, prediction, and learning. Our residual MLP is a coarse instantiation of the forward model in that picture; the multi-step supervision we add is motivated by the fact that the loop that consumes the estimate operates over hundreds of integration steps and thus depends on long-horizon prediction quality. More biologically grounded forward-model families — continuous-time recurrent networks with input-modulated time constants such as liquid time-constant networks [10] and their closed-form variant [11] — have been proposed for sensorimotor dynamics; in this work we keep the forward-model implementation intentionally generic to isolate the oracle-choice question.

Learned dynamics and multi-step supervision. Model-based reinforcement-learning systems have moved from one-step dynamics losses to multi-step rollout objectives because closed-loop planners need predictions that survive being unrolled. Dreamer [12] backpropagates through imagined trajectories in a latent world model, and MBPO [13] branches short model rollouts from real data, formalising "when to trust your model". We take the same lesson into observer design: a forward model that is accurate enough over the controller's planning horizon shifts where prediction-error correction is helpful and where it is harmful.

Outcome-driven adaptation and SPSA. Outcome-driven, gradient-free adaptation has a long history in motor control: trial-by-trial visuomotor adaptation in humans [2] is captured by an internal-model update rule that uses only the observed end-of-trial error, not the underlying optimal-gain calculation. Algorithmically, our across-trial loop follows Spall's Simultaneous Perturbation Stochastic Approximation (SPSA) [6]: a two-point gradient estimate using a single Rademacher perturbation per iteration, with a robust hyperparameter schedule. SPSA's appeal in this setting is that it needs only the per-episode outcome, not the per-step state, and that its computational cost scales with the number of meta-parameters β (here, 10) rather than the dimension of the observer state (here, 83). Behaviour cloning and DAgger [14], [15] were used in an earlier version of this work to ask whether the resulting controllers consume state estimates; we found that with small demonstration sets they do not (Appendix C), and we now use only state-coupled feedback controllers in the main results so the observer signal flows to muscle commands.

Muscle-driven simulation. Our task runs in MuJoCo [8] via the MyoSuite musculoskeletal benchmark [7], which provides a 34-muscle reaching arm and a contact-rich set of tasks suitable for sensorimotor estimation studies. MyoSuite is increasingly used as a testbed for reinforcement-learning controllers; here we use it instead to ask whether a predictive state observer can adapt its correction-gain from agent-available signals alone, a question that the RL-centric literature has not addressed.

III. Method

A. Task and Environment

We use the `myoArmReachFixed-v0` task in MyoSuite 2.12.2, a 34-muscle shoulder-elbow-wrist-hand system actuated by activations in $[0,1]^{34}$. The task is to drive the index-finger tip to a randomly drawn target in a 12 s episode at Δt=0.02 s (600 control steps). The flat state $x \in \mathbb{R}^{83}$ stacks joint positions $q \in \mathbb{R}^{20}$, joint velocities $\dot{q} \in \mathbb{R}^{20}$, muscle activations $a \in \mathbb{R}^{34}$, finger-tip position $p_{\text{tip}} \in \mathbb{R}^3$, target $p_{\text{tgt}} \in \mathbb{R}^3$, and reach error $e = p_{\text{tgt}} - p_{\text{tip}} \in \mathbb{R}^3$. True state is extracted from MuJoCo and used only as an oracle for evaluation; the closed loop never accesses it (Fig. 1).

B. Forward Model

A residual MLP $f_\theta(x_t, u_t)$ predicts the state delta $\Delta x_t = x_{t+1} - x_t$ rather than x_{t+1} directly. At Δt=0.02 s the state changes slowly, so $x_{t+1} \approx x_t$ component-wise; predicting Δx frees the model from re-learning the identity map and concentrates capacity on the small per-step dynamics that the filter and controller actually consume. The parameterisation is standard in model-based RL world models [12], [13]. The input is the flat 83-dim state and the 34-dim excitation; the architecture is two hidden layers of 256 units with ReLU and LayerNorm; training uses Adam [16]. In the single-step baseline, f_θ is trained on the standard one-step MSE loss

$$\mathcal{L}_1(\theta) = \mathbb{E}_{(x_t, u_t, x_{t+1})}\left[\|f_\theta(x_t, u_t) - \Delta x_t\|^2\right]. \tag{1}$$

Multi-step rollout supervision. The multi-step variant trains f_θ with an H-step rollout loss instead of the one-step MSE, where H denotes the training rollout horizon (distinct from the correction gain K defined in Sec. III-C). For each contiguous trajectory chunk $(x_t, u_t, \ldots, x_{t+H})$ inside a single source episode, the model is unrolled free-running: $\hat{x}_{t+k} = \hat{x}_{t+k-1} + f_\theta(\hat{x}_{t+k-1}, u_{t+k-1})$, $\hat{x}_t = x_t$, and the loss averages MSE across the H prediction steps:

$$\mathcal{L}_H(\theta) = \mathbb{E}\Big[\tfrac{1}{H}\sum_{k=1}^{H}\|\hat{x}_{t+k} - x_{t+k}\|^2\Big]. \tag{2}$$

Multi-step rollout supervision has been used to stabilise long-horizon predictions in latent-imagination world models [12] and in model-based policy optimisation [13], where the relevant horizon is the planner's rollout depth rather than the observer's delay; we re-use the technique here to make f_θ robust across the observer's buffer rollout.
We train the same architecture with $H \in \{1, 4, 8\}$ on the same expanded dataset (51 k transitions across 106 episodes; see Sec. IV-A).

C. Fixed-Lag Predictive State Observer

The observer uses an innovation-update form structurally familiar from Kalman filtering but does not propagate covariances and does not compute an optimal Kalman gain. The gain $K \in [0,1]$ is a *sensory prediction-error correction gain*, not a covariance-derived Kalman gain.

The observation pipeline produces $y_t = h(x_{t-d}) + \eta_t$ where h is the identity (the state vector and the observation share the

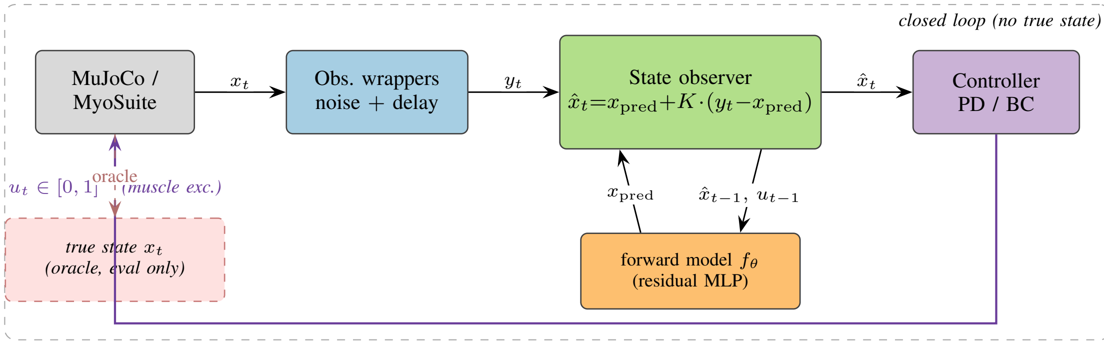

Fig. 1. **Closed-loop pipeline.** Each control step, the env's true state x_t is filtered through observation wrappers that add per-field Gaussian noise and a fixed delay, producing y_t. The predictive state observer combines y_t with a one-step prediction x_{pred} produced by the residual-MLP forward model $f_\theta(\hat{x}_{t-1}, u_{t-1})$ via a scalar correction gain $K \in [0,1]$ on the sensory prediction error $y_t - x_{\text{pred}}$ (Eq. 4). The controller consumes only the observer output $\hat{x}_t$ and emits the next muscle excitation $u_t \in [0,1]^{34}$. The true state x_t is exposed only to the offline evaluator (dashed red box); the closed loop never accesses it.

same schema), $d \in \mathbb{Z}_{\geq 0}$ is a fixed delay in control steps, and η_t is per-field i.i.d. Gaussian noise with diagonal covariance $\operatorname{diag}(\sigma^2)$. The estimator maintains a length-$(d+1)$ ring buffer of past one-step estimates $(\hat{x}_{t-d}, \ldots, \hat{x}_{t-1})$ together with the corresponding applied actions $(u_{t-d}, \ldots, u_{t-1})$. At each step the observer first runs the forward model on every buffered estimate to obtain a prediction at the present time,

$$\tilde{x}_t = \hat{x}_{t-1} + f_\theta(\hat{x}_{t-1}, u_{t-1}), \tag{3}$$

then forms an innovation against the *delayed* buffer entry and applies an element-wise gain $K \in [0,1]^{83}$:

$$\hat{x}_{t-d} \leftarrow \hat{x}_{t-d} + K \odot \big(y_t - \hat{x}_{t-d}\big), \tag{4}$$

and finally re-rolls the corrected past estimate forward through the buffered actions to obtain the present-time estimate $\hat{x}_t$. Eq. (4) collapses to the standard one-step innovation update when $d{=}0$. Throughout the paper K is a scalar broadcast across the 83 fields so the correction gain reduces to a single trust knob between forward prediction ($K{=}0$) and observation ($K{=}1$). This fixed-lag construction follows the buffered Rauch–Tung–Striebel form [5], [9] specialised to a scalar gain.

D. Reliability-Adaptive Correction Gain (Within-Trial)

The observer above uses a fixed scalar K broadcast across all 83 fields. We propose instead an *agent-available, field-wise reliability-adaptive* rule that sets $K_f(t)$ online from each sensory channel's own innovation history. No oracle on the optimal gain, no true state, and no swept-condition labels enter the rule; only the per-step innovation $e_t = y_t - \hat{x}_{t-d}$ does.

We partition the 83-dim state into five *sensory fields*: qpos (20), qvel (20), act (34), tip_pos (3), and reach_err (3). The partition is an engineering grouping driven by the MyoSuite state schema, but it admits a coarse biological reading: qpos and qvel are proprioceptive-like, act is efferent / muscle-command-like, and tip_pos and reach_err are visual / task-space-like. The target-position channel (3) is held at a fixed gain because the target is known and does not have a meaningful prediction error. For each field $f \in \mathcal{F} = \{\text{qpos}, \text{qvel}, \text{act}, \text{tip\_pos}, \text{reach\_err}\}$ the observer maintains an exponentially-weighted (EMA) estimate of the per-field squared-innovation mean,

$$v_f(t) = (1-\alpha)\, v_f(t-1) + \alpha \, \frac{1}{|\mathcal{I}_f|} \sum_{i \in \mathcal{I}_f} e_{t,i}^2, \tag{5}$$

where $\mathcal{I}_f$ are the field's coordinate indices in the 83-dim state vector and $\alpha{=}0.05$ unless otherwise noted. The agent's running estimate of *sensory reliability* for field f is the inverse of this variance,

$$r_f(t) = \frac{1}{\varepsilon + v_f(t)}, \tag{6}$$

with $\varepsilon{=}10^{-6}$. We then map reliability to a per-field correction gain through a two-parameter logistic,

$$K_f(t) = \sigma\big(\beta_{0,f} + \beta_{1,f}\, \log r_f(t)\big), \quad \sigma(z) = \frac{1}{1+e^{-z}}, \tag{7}$$

and form the 83-dim gain vector $K(t) \in [0,1]^{83}$ by broadcasting $K_f(t)$ over its field's coordinates. The innovation update Eq. (4) is then applied elementwise with $K(t)$ in place of the scalar K.

Eq. (7) is intentionally simple. The slope $\beta_{1,f}$ controls how sharply the gain reacts to changes in log-reliability, and the intercept $\beta_{0,f}$ controls the operating point at $r_f(t) = 1$ (i.e., $K_f = \sigma(\beta_{0,f})$). With $\beta_{0,f} = 0$, $\beta_{1,f} = 0.5$, and $v_f(0) = 1$, the rule produces $K_f(t) \approx 0.5$ on the first step and adapts upward when innovation shrinks (the sensor is informative) or downward when innovation grows (the sensor is noisy). Per-field $\beta = \{\beta_{0,f}, \beta_{1,f}\}$ lets the rule learn that different fields have different reliability–correction trade-offs under the same physical noise level.

E. Across-Trial Outcome Adaptation (SPSA)

The meta-parameters $\beta = \{(\beta_{0,f}, \beta_{1,f}) : f \in \mathcal{F}\} \in \mathbb{R}^{10}$ themselves can be adapted across trials from *outcome* signals the agent has at the end of each episode. We define the per-episode outcome as the minimum tip-to-target distance,

$$\operatorname{minTip}(\beta;\, \sigma, d) \;=\; \min_{t \in [0,T]} \|p_{\text{tip}}(t) - p_{\text{tgt}}\|, \tag{8}$$

which the agent observes by proprioception and visual feedback after the reach. We use minTip as an *operational scalar* *proxy* for task outcome — not as a claim that the nervous system observes this exact variable. The across-trial loop minimises $\mathbb{E}[\mathrm{minTip}(\beta)]$ where the expectation is over target draws (and, in the multi-cell variant, over uniformly drawn (σ, d) cells from the focused grid).

Because the closed-loop minTip is not differentiable in β (it passes through env steps, EMA state, and a non-smooth min), we use Simultaneous Perturbation Stochastic Approximation (SPSA) [6]: at iteration n, draw a Rademacher vector $\Delta_n \in \{\pm 1\}^{10}$, evaluate the outcome at $\beta_n \pm c_n \Delta_n$ averaged over S paired episodes (samples per side), form the central-difference gradient estimate

$$\hat{g}_n = \frac{\mathrm{minTip}(\beta_n + c_n\Delta_n) - \mathrm{minTip}(\beta_n - c_n\Delta_n)}{2c_n}\,\Delta_n^{-1}, \tag{9}$$

and update $\beta_{n+1} = \beta_n - a_n \hat{g}_n$. The Spall schedules are $a_n = a/(n{+}1{+}A)^{\alpha_s}$ and $c_n = c/(n{+}1)^{\gamma_s}$ with a=2.0, c=0.3, α_s=0.602, γ_s=0.101, and A=5. Two SPSA variants are reported: *single-cell* (S=12, 100 iterations, σ=none, d=18) and *full-grid* (S=12, 100 iterations, uniformly drawn (σ, d) from the 6-cell grid per iteration).

a) Comparison with oracle-supervised gain.: For a diagnostic upper bound on what *any* K-setting strategy can achieve on this task, Appendix C reports an offline-oracle-supervised correction-gain predictor that trains a small MLP $g_\phi(\sigma, d, c) \to [0, 1]$ against either an open-loop estimation-accuracy oracle $K^\star_{\mathrm{ol}}(\sigma, d, c)$ or a closed-loop task-error oracle $K^\star_{\mathrm{cl}}(\sigma, d, c) = \arg\min_K \mathbb{E}[\mathrm{minTip}(K) \,|\, \sigma, d, c]$. That predictor requires access to a swept K-grid the biological agent does not have; we include it strictly as the strongest non-agent-available baseline.

F. Controllers

Three controllers are evaluated as closed-loop wrappers around the estimator.

(i) Heuristic projection. A fixed random projection $W \in \mathbb{R}^{34\times 3}$ (drawn once with a fixed seed from $\mathcal{N}(0,1)^{34\times 3}$) maps the estimated 3-dim reach-error $\hat{r}$ to a 34-dim muscle excitation through an element-wise logistic $\mathrm{sigm}(\cdot)$ (distinct from the noise scale vector σ above):

$$u = \mathrm{sigm}(b - g\,W\,\hat{r}), \quad b{=}0,\ g{=}5, \tag{10}$$

producing a state-sensitive baseline that does not actually reach.

(ii) Joint-space PD with inverse kinematics. For each target tip position $p^\star \in \mathbb{R}^3$ we solve $q^\star = \mathrm{IK}(p^\star)$ once at episode start via a damped-Jacobian Newton iteration

$$q^{(k+1)} = q^{(k)} + J^\top \big(JJ^\top + \lambda^2 I\big)^{-1}\big(p^\star - p(q^{(k)})\big), \tag{11}$$

where $J = \partial p/\partial q$ is the tip-site position Jacobian obtained from MuJoCo's `mj_jacSite` and λ^2=0.1 is the damping. The arm is then driven by a joint-space PD law and a moment-arm muscle map:

$$u_{\text{joint}} = K_p\,(q^\star - \hat{q}) - K_d\,\hat{\dot{q}}, \tag{12}$$

$$u_{\text{muscle}} = \mathrm{clip}(\alpha\,\mathrm{ReLU}(M\,u_{\text{joint}}),\ 0,\ 1), \tag{13}$$

where $M \in \mathbb{R}^{34\times 20}$ is the muscle-tendon moment-arm matrix (the dense form of MuJoCo's sparse `actuator_moment` at the env's current pose), and $\alpha = 5$ is a scalar action scale. Tuned gains: $K_p{=}30$, $K_d{=}3$.

(iii) Endpoint-error sensorimotor policy. A task-level analogue of (ii) that skips the IK pre-step and maps the estimated tip-to-target error to muscle excitation via a Cartesian PD law projected through the tip Jacobian and the moment-arm matrix:

$$u_{\text{tip}} = K_{p,e}\,(p^\star - \hat{p}_{\text{tip}}) - K_{d,e}\,J\hat{\dot{q}}, \tag{14}$$

$$u_{\text{joint}} = J^\top\,u_{\text{tip}}, \tag{15}$$

$$u_{\text{muscle}} = \mathrm{clip}\big(\alpha\,\mathrm{ReLU}(M\,u_{\text{joint}}),\ 0,\ 1\big), \tag{16}$$

with $K_{p,e}$=30, $K_{d,e}$=3, α=5 and J the tip-site positional Jacobian captured at the env's neutral pose. We use this as a transparent task-level state-coupled policy that lets the observer output flow directly to muscle commands without an IK pre-step; the $J^\top$-style projection is the classic Jacobian-transpose method and is *not* an optimal feedback controller (no Riccati / LQR / OFC is solved here).

(iv) Behaviour-cloned MLP (diagnostic only). A two-hidden-layer (256-unit) sigmoid policy trained on 30 scripted IK demonstrations [14] (no DAgger [15]). We retain this controller only as a diagnostic in Appendix C to test whether a learned downstream policy consumes the observer signal; the main results of Sec. IV-B–IV-E use joint-PD (ii) so the observer output flows to muscle commands through an explicit feedback law.

G. Evaluation Pipeline

Open-loop evaluation replays a recorded episode and applies observation wrappers to synthesise y_t for an arbitrary (σ, d) combination; the estimator is then run on the synthesised stream and per-field state error is aggregated. *Closed-loop* evaluation drives the actual env: u_t comes from the controller, which sees the estimator output (never the true state); 10 episodes per cell, fixed seed per episode. The same evaluation grid (3 controllers × 6 noise levels × 4 delays = 72 open-loop conditions; the focused closed-loop grid is 3 noise levels × 2 delays × 10 episodes = 60 episodes per estimator) is reused across phases.

IV. Results

A. Forward-model improvement cycle

An initial forward model trained on the original ∼ 3.6 k transitions from 6 episodes had low single-step error but unbounded long-horizon rollouts (h=50 MSE ∼ 10.6). Collecting 100 additional `random` and `lowamp` episodes (a total of 51 k transitions across 106 episodes) and retraining the same architecture cut h=50 MSE to 0.052 — a ∼ 200× improvement. We use this expanded dataset, and the resulting model, as the single-step baseline throughout the paper.

B. Default within-trial reliability is task-misaligned (C1)

We first evaluate the reliability-adaptive observer with a deliberately neutral default β: $\beta_{0,f} = 0$, $\beta_{1,f} = 0.5$ for all five fields, $\alpha = 0.05$, $v_f(0) = 1$. At iteration 0 this produces $K_f \approx \sigma(0) = 0.5$; as the EMA tracks each field's running squared innovation, K_f adapts upward when the channel becomes informative and downward when it becomes noisy.

Across the focused 3 noise $\times$ 2 delay grid under the H=8 forward model, the realised per-field K_f averaged over the second half of a representative episode is highly heterogeneous ($K_f \in [0.31, 0.98]$ across the grid; the per-field cell means are reported in Table I). qvel is the *most-suppressed* channel (cell range 0.31–0.49, mean 0.39), reflecting bounded but informative velocity innovation; qpos, act, tip_pos, and reach_err settle in the 0.59–0.98 range, close to K=1 at delay-0 cells where the channel innovation collapses to near-zero. Closed-loop min-tip averages 0.755 m across the six cells. For comparison, K=0 averages 0.495 m and K=1 averages 0.732 m on the same grid (Fig. 2).

TABLE I
REALISED PER-FIELD CORRECTION GAIN K_f UNDER DEFAULT WITHIN-TRIAL RELIABILITY ($\beta_{0,f}$=0, $\beta_{1,f}$=0.5). EACH ENTRY IS THE SECOND-HALF MEAN OF A SINGLE REPRESENTATIVE EPISODE ON THE FOCUSED GRID (TARGET INDEX 0, H=8 FORWARD MODEL, JOINT-PD CONTROLLER). QVEL IS THE ONLY FIELD THAT THE DEFAULT RULE DEEPLY SUPPRESSES; THE OTHER FOUR SIT CLOSE TO K=1 AT DELAY-0 CELLS.

noise	delay	qpos	qvel	act	tip_pos	reach_err
none	0	0.96	0.49	0.91	0.97	0.98
none	18	0.70	0.31	0.69	0.59	0.68
high	0	0.96	0.49	0.92	0.96	0.97
high	18	0.69	0.32	0.71	0.68	0.69
xhigh	0	0.90	0.45	0.91	0.94	0.94
xhigh	18	0.67	0.31	0.69	0.64	0.68

The structural reason is that under the H=8 forward model the closed-loop oracle on this task is $K^\star_{\rm cl} = 0$ in every cell of the focused grid (Appendix E): the forward model is accurate enough over the controller's planning horizon that any observation correction shifts the trajectory away from the planned IK target. Reliability-based adaptation *by construction* cannot produce $K_f = 0$ unless either $\beta_{0,f} \to -\infty$ or the EMA variance grows without bound; under bounded innovation neither happens. The default rule therefore trails K=0 by 0.23–0.29 m in min-tip across the grid. We read this as a fundamental signal: *innovation reliability alone, without an outcome objective that knows the task, cannot recover the right correction-gain level when the forward model is good enough to operate the loop without sensory help.*

C. Single-cell SPSA recovers $K^\star_{\rm cl}$-equivalent performance (C2)

We next add the across-trial outcome loop of Sec. III-E. We run SPSA for 100 iterations on the single (σ=none, d=18) cell with $S = 12$ samples per side and the Spall schedule of Sec. III-E. The outcome objective is the per-episode min-tip averaged across the S paired episodes at the two perturbed β points.

Figure 3 reports the outcome trajectory and the final-iteration per-field β. The per-iteration outcome trajectory falls from $\approx$0.61 m at iteration 0 to a last-20-iter mean of $\approx$0.49 m,

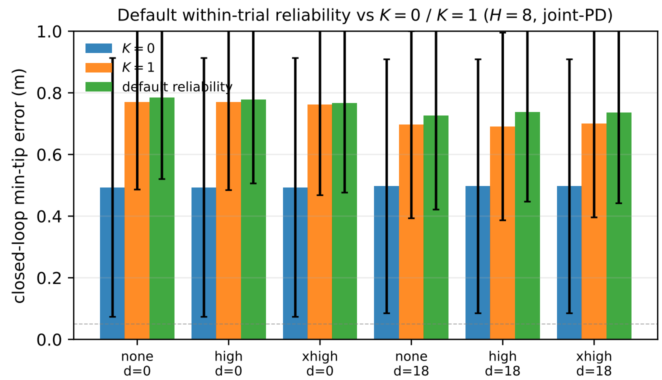

Fig. 2. **Closed-loop min-tip error under the H=8 forward model and joint-PD controller for three estimators on the focused 3 noise $\times$ 2 delay grid (mean $\pm$ std over 10 episodes per cell).** The default reliability-adaptive rule ($\beta_{0,f} = 0$, $\beta_{1,f} = 0.5$, all fields) settles to $K_f \approx 0.55$–0.72 and trails K=0 by 23–29 cm across cells; K=1 trails K=0 by 14–28 cm. The closed-loop oracle $K^\star_{\rm cl}$ on this grid is K=0 at every cell (Appendix E); a reliability rule cannot produce K_f=0 from bounded innovation alone.

within 1 cm of the K=0 baseline on this cell (0.50 m, from Appendix E); the single iteration-99 snapshot is noisier at 0.53 m, and a fresh deployed evaluation of the final β on 10 independent episodes gives 0.56 m (residual $\sim$ 6 cm to K=0). The final β is non-uniform across fields: qpos becomes simultaneously the most-suppressed ($\beta_{0,\rm qpos} \approx -1.53$) and the most-reactive ($\beta_{1,\rm qpos} \approx +0.89$) field, while the remaining fields cluster at moderate values ($\beta_{0,f} \in [-0.35, -0.04]$, $\beta_{1,f} \in [-0.02, +0.50]$). reach_err's slope flattens to near zero, indicating SPSA found no benefit from making the reach-error channel innovation-reactive. Field specialisation emerges directly from outcome feedback, with no per-field oracle.

The key implication is that the agent does not need an oracle on $K^\star_{\rm cl}$: from per-episode reach outcomes alone, 10-parameter SPSA on β recovers about 17 of the 23 cm default-reliability vs K=0 gap on a single training cell (0.77 m $\to$ 0.56 m deployed, against K=0 at 0.50 m), with the smoothed training trajectory briefly reaching the K=0 level.

D. Multi-cell SPSA is bottlenecked by cell heterogeneity (C3)

A single-cell-trained β is task-specific. To test whether a single β can serve the whole focused grid, we repeat the SPSA loop with each iteration's $S = 12$ paired episodes drawn uniformly across the 6 cells. The outcome objective is the 6-cell mean min-tip.

After 100 iterations the multi-cell β converges to β_0 = {qpos:-0.64, qvel:-0.21, act:0.12, tip_pos:-0.32, reach_err:-0.16}, β_1 = {qpos:0.20, qvel:0.80, act:0.54, tip_pos:0.63, reach_err:0.33}, with $K_{\rm mean}$ averaged across fields equal to $\approx$0.51 at delay-18 cells and $\approx$0.63 at delay-0 cells. The realised closed-loop min-tip improves over default reliability by 3–5 cm at the three delay-18 cells and by $\sim$0 cm at the three delay-0 cells. Compared to K=0, the multi-cell β still trails by 19–22 cm at delay-18 cells and by $\sim$28 cm at delay-0 cells.

The structural reason is that $K^\star_{\rm cl} = 0$ uniformly across the focused grid (Appendix E), yet field-wise reliability under

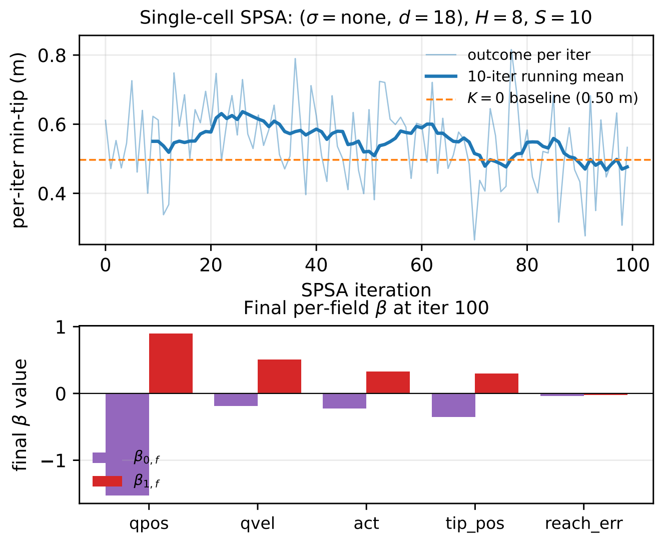

Fig. 3. **Single-cell SPSA on (σ=none, d=18) under the H=8 forward model and joint-PD controller.** *Top*: per-iteration min-tip outcome (mean over the S=10 paired episodes per side); the dashed line marks the closed-loop K=0 baseline on this cell (0.50 m). The per-iter outcome falls from $\approx$0.61 m at iteration 0 to a last-20-iter mean of $\approx$0.49 m, reaching the K=0 baseline in the smoothed training-time signal; a fresh deployed evaluation of the final β gives 0.56 m (residual $\sim$6 cm to K=0). *Bottom*: final per-field β vector. qpos has both the most-negative intercept (-1.5) and the steepest slope ($+0.9$); the resulting per-field correction gains K_f specialise.

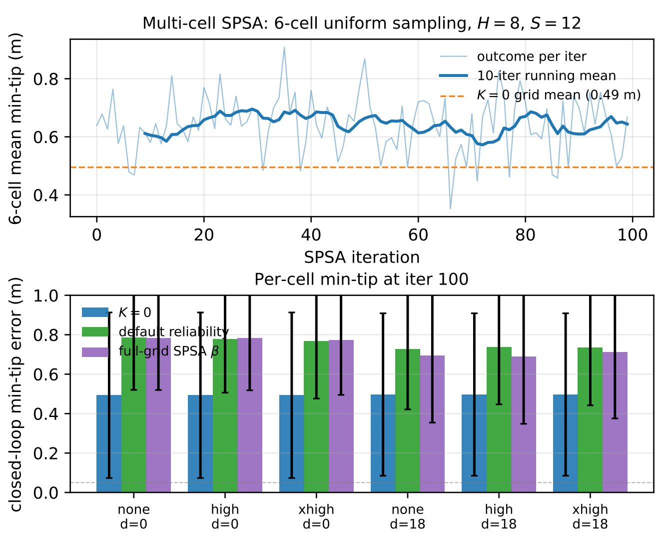

Fig. 4. **Multi-cell SPSA over the focused 6-cell grid under the H=8 forward model and joint-PD controller.** *Top*: per-iteration 6-cell-mean min-tip outcome (mean over S=12 paired draws). *Bottom*: per-cell min-tip at iteration 100 for K=0, default-reliability, and the multi-cell-trained β. Improvement over default reliability is 3–5 cm at delay-18 cells and $\sim$0 cm at delay-0 cells; the multi-cell β still trails K=0 by 19–22 cm at delay-18 cells.

H=8 has different absolute levels and time courses at each cell. A single β would need $\beta_{0,f}$ deeply negative (a few standard deviations below the realised $\log r_f(t)$ range) to drive every cell's K_f near 0; SPSA does not reach this regime in 100 iterations because each intermediate β_0 value is already differentiable improvement in some cells and worse in others. The cell-heterogeneity bottleneck is therefore not a failure of outcome adaptation itself — the single-cell run of C2 converged — but evidence that the parameterisation, a single global $\beta \in \mathbb{R}^{10}$, is underpowered when the reliability geometry varies across cells. A natural next model is a context-conditioned β network that maps a short sliding window of field-wise innovation statistics (mean, variance, autocorrelation) to the ten β parameters, optionally with known (σ, d) context as an ablation-only auxiliary input; the agent-available primary input is the innovation window, since a biological agent does not directly observe the noise level or sensorimotor delay.

E. Task transfer reveals task-dependence (C4)

The previous results all use ReachFixed, where the same target location is reused across episodes. We test the same observer on the *variable-target* variant `myoArmReachRandom-v0`, where the target is freshly sampled per episode from the workspace distribution. The forward model for this transfer slice is also trained on ReachRandom rollouts (Appendix F) so the forward-model accuracy axis is matched.

TABLE II
TRANSFER OF THE RELIABILITY-ADAPTIVE OBSERVER TO REACHRANDOM (VARIABLE TARGET). MEAN min-TIP (M) OVER 10 EPISODES PER CELL UNDER THE H=8 REACHRANDOM FORWARD MODEL AND JOINT-PD CONTROLLER. "DEFAULT" IS THE $\beta_{0,f}$=0, $\beta_{1,f}$=0.5 RULE; "TRANSFERRED" IS THE REACHFIXED FULL-GRID-SPSA-LEARNED β FROM C3 APPLIED WITHOUT RETRAINING.

noise	delay	K=0	K=1	default	transferred
none	0	0.670	0.770	0.781	0.768
none	18	0.642	0.647	0.631	0.680
high	0	0.670	0.770	0.776	0.768
high	18	0.642	0.639	0.608	0.692
xhigh	0	0.670	0.762	0.771	0.759
xhigh	18	0.642	0.627	0.649	0.686

Table II reports the result. On ReachRandom the closed-loop oracle $K^\star_{\text{cl}}$ takes three distinct values across the grid (Appendix F, Table X): K=0 at the three delay-0 cells, K=0.25 at (none, d=18) and (high, d=18), and K=1 at (xhigh, d=18). At the delay-18 cells, the default reliability rule *beats* K=0 at 2 of 3 cells (by 1.1 cm at (none, d=18) and 3.4 cm at (high, d=18)); at (xhigh, d=18) it loses by 0.7 cm. The transferred ReachFixed-trained β, in contrast, trails K=0 at every delay-18 cell by 3.8–5.0 cm.

Two findings combine here. *First*, on a task whose $K^\star_{\text{cl}}$ is heterogeneous, agent-available default reliability is already competitive — no outcome adaptation needed. *Second*, β trained on a different task does not transfer when the source task's $K^\star_{\text{cl}}$ is uniform but the target task's is heterogeneous: the source task pushes β in a direction that is harmful on the target. This is direct evidence that the correction-gain rule must be sensitive to the *distribution of the closed-loop optimum*, not merely to the noise and delay regime that within-trial reliability sees.

F. Comparison with offline-oracle-supervised gain

For a diagnostic upper bound we compare against an offline-oracle-supervised correction-gain predictor (Appendix C): a small MLP $g_\phi(\sigma, d, c)$ trained against the open-loop oracle $K^\star_{\mathrm{ol}}$ and, separately, against the closed-loop oracle $K^\star_{\mathrm{cl}}$ obtained by a per-cell K-sweep. On the focused grid under $H{=}8$ the closed-loop-oracle variant outputs $K \approx 0.02$ and matches the $K{=}0$ baseline within 1.5 cm at all delay-18 cells; the open-loop-oracle variant outputs $K \approx 1$ and trails $K{=}0$ by 14–28 cm. The closed-loop-oracle variant is therefore the strongest non-agent-available baseline available in this setting, and the C2 single-cell SPSA result effectively matches it on its training cell using only agent-available signals.

V. Discussion

Innovation alone is not enough; outcome closes the gap. The C1 finding is the central structural observation: a within-trial reliability rule that uses only the per-channel innovation history settles on $K_f \approx 0.6$–0.7 across the focused grid, while the closed-loop oracle on the same grid is $K^\star_{\mathrm{cl}} = 0$ everywhere. The gap is not an SPSA failure or a meta-parameter mis-tuning; it is structural. Innovation gives the agent information about *sensor quality*, not about *whether the forward model is good enough to operate without sensory help*. The second piece of information — whether the controller's planned trajectory is being followed — is in the task outcome, not in the innovation. C2 then shows that adding the outcome layer through SPSA recovers about 17 of the 23 cm default-reliability gap on a single training cell (0.77 m $\to$ 0.56 m deployed, against $K{=}0$ at 0.50 m); the smoothed training trajectory briefly reaches the $K{=}0$ level as a convergence signal. Together they show that, at least on this task, a two-layer agent-available adaptation rule is sufficient. Open question: whether reliability + outcome can also be combined within a single trial (e.g. an outcome-modulated β during reaching) rather than only across trials.

Single-β adaptation has a structural ceiling under cell heterogeneity. The C3 result — multi-cell SPSA closes only 3–5 cm of the ~ 29 cm gap at delay-18 cells, and none at delay-0 cells — should not be read as a failure of outcome adaptation itself. The single-cell C2 run converged, and the structural reason for the multi-cell ceiling is geometric: under $H{=}8$ on ReachFixed, $K^\star_{\mathrm{cl}} = 0$ at every cell, but the realised reliability $r_f(t)$ traces differ substantially across cells (Table I). A single global $\beta_{0,f}$ that pushes K to nearly 0 at the high-noise cell over-suppresses at the low-noise cell and vice versa. C3 therefore evidences that the *adaptation parameterisation*, not the outcome loop, is underpowered. The natural next model class is a context-conditioned β network mapping a sliding window of field-wise innovation statistics to the ten β parameters (known (σ, d) context being available only as an ablation-only auxiliary input, since the biological agent does not observe these labels directly); corresponding biologically to context-dependent gating. We have not implemented it here. A related theoretical question is whether $K^\star_{\mathrm{cl}}$ under reasonable controllers admits a low-dimensional shared structure across cells; if it does, a context-dependent extension of β should suffice.

Task transfer reveals that the right rule depends on $K^\star_{\mathrm{cl}}$ geometry. The two C4 findings combine as follows. *First*, default within-trial reliability is competitive with $K{=}0$ on ReachRandom and slightly better at 2 of 3 delay-18 cells (by 1–3 cm); *second*, the ReachFixed-trained β trails $K{=}0$ at every delay-18 cell by ~ 4–5 cm. The mechanism is that $K^\star_{\mathrm{cl}}$ on ReachRandom is multimodal across cells ($\{0, 0.25, 1\}$, Appendix F), so the default field-wise gains happen to land near the per-cell optima in the delay-18 rows. Pushing K_f globally down (as the ReachFixed-trained β does) is correct on the source task but wrong on the target. The reading is that gain adaptation trained in a uniform-optimum regime should transfer poorly to a task whose closed-loop optimum is multimodal unless the rule is context-conditioned — a testable computational hypothesis that the present transfer slice illustrates within a single simulated task family.

Oracle-supervised gain as diagnostic upper bound. Appendix C reports an oracle-supervised correction-gain predictor that requires per-cell K-sweep labels. The closed-loop-oracle variant outputs $K \approx 0.02$ and matches $K{=}0$ on the focused grid within 1.5 cm at delay-18 cells, which is the strongest non-agent-available baseline for the correction-gain problem. The C2 single-cell SPSA result effectively reaches this level using only signals the agent has access to — prediction errors during the reach and the reaching outcome afterward. We frame the oracle-supervised predictor not as a biological model but as the upper bound against which agent-available adaptation should be measured. The fact that a 10-parameter SPSA on per-episode outcome closes the gap on a single training cell suggests the upper bound is reachable by plausible biological signals; whether it is reachable at the population level (i.e. a single β over many cells) appears to depend on $K^\star_{\mathrm{cl}}$ structure as C3 shows.

Limits and open questions. (a) Two-layer rule only: β is the only learned object inside the reliability-adaptive observer; the forward model is fixed. A jointly-trained forward model + β pair is a natural extension. (b) Single-β SPSA: the multi-cell C3 result is a ceiling of single-global-β adaptation; a context-dependent β network is the obvious next variant and is left to future work. (c) Two MyoSuite reach variants (ReachFixed and ReachRandom), 34-muscle arm only. Transfer to contact-rich tasks (grasp/place) and to other MyoSuite arms (Elbow/Finger reach) remains open. (d) Forward-model family: the residual MLP is a coarse instantiation of the cerebellar forward-model box; continuous-time recurrent alternatives such as LTC [10] and CfC [11] would provide a closer computational analogue when this observer is plugged into a full cortico-cerebellar loop. (e) The behaviour-cloned controller (Appendix C) washes out estimator differences at the demonstration scale we test; closed-loop state-estimation benchmarks should keep this in mind when choosing the downstream policy. (f) The oracle-supervised predictor of Appendix C is intentionally excluded from the main adaptive observer because it requires per-condition K-sweep labels the agent cannot generate online; it is reported only as a non-agent-available upper bound against which the within-trial + outcome adaptation of C2 can be measured.

VI. Conclusion

We have shown on a muscle-driven reaching task that a forward-model-based predictive state observer can adapt its sensory prediction-error correction gain from signals the agent itself generates online — within-trial innovation reliability and across-trial reaching outcome — rather than from the per-condition oracle labels that supervised gain predictors require at training time (Appendix C). A two-layer adaptation rule combining a logistic $K_f(t)=\sigma(\beta_{0,f}+\beta_{1,f}\log r_f(t))$ over field-wise reliability with SPSA across-trial updates of β recovers about 17 of the 23 cm default-reliability vs $K=0$ gap on a single training cell (0.77 m $\rightarrow$ 0.56 m deployed against $K=0$ at 0.50 m; the smoothed training trajectory briefly reaches the $K=0$ level as a convergence signal), without per-condition oracle supervision (C2). The same rule has two structural limitations: when the closed-loop optimum is uniform across noise-delay cells, a single global β trained on a multi-cell mixture closes only 3–5 cm of the gap (C3, cell-heterogeneity ceiling); and a β trained on one task does not transfer to a task whose closed-loop optimum is multimodal (C4, task-dependent rule). We read these as direct evidence that the correction-gain problem is solvable from agent-available signals but that the resulting rule must be context-dependent, not a single global β. Within the forward-model + predictive-state-observer framing this is a model of how a biological motor system might assemble its correction-gain rule from prediction errors and reaching outcomes without ever seeing the oracle.

Acknowledgement

The author thanks the MyoSuite maintainers for releasing the musculoskeletal benchmark used in this study.

References

[1] N. A. Bernstein, *The Coordination and Regulation of Movements*. Oxford: Pergamon Press, 1967.
[2] D. M. Wolpert and Z. Ghahramani, "Computational principles of movement neuroscience," *Nature Neuroscience*, vol. 3, no. 11, pp. 1212–1217, Nov. 2000.
[3] R. E. Kalman, "A new approach to linear filtering and prediction problems," *Transactions of the ASME–Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, Mar. 1960.
[4] R. K. Mehra, "On the identification of variances and adaptive Kalman filtering," *IEEE Transactions on Automatic Control*, vol. 15, no. 2, pp. 175–184, Apr. 1970.
[5] B. D. O. Anderson and J. B. Moore, *Optimal Filtering*. Englewood Cliffs, NJ: Prentice-Hall, 1979, republished by Dover, 2005.
[6] J. C. Spall, "Multivariate stochastic approximation using a simultaneous perturbation gradient approximation," *IEEE Transactions on Automatic Control*, vol. 37, no. 3, pp. 332–341, 1992.
[7] V. Caggiano, H. Wang, G. Durandau, M. Sartori, and V. Kumar, "MyoSuite – a contact-rich simulation suite for musculoskeletal motor control," in *Proc. 4th Annual Learning for Dynamics and Control Conference (L4DC)*, ser. Proceedings of Machine Learning Research, vol. 168. PMLR, 2022. [Online]. Available: https://proceedings.mlr.press/v168/caggiano22a.html
[8] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2012, pp. 5026–5033.
[9] H. E. Rauch, F. Tung, and C. T. Striebel, "Maximum likelihood estimates of linear dynamic systems," *AIAA Journal*, vol. 3, no. 8, pp. 1445–1450, Aug. 1965.
[10] R. Hasani, M. Lechner, A. Amini, D. Rus, and R. Grosu, "Liquid time-constant networks," in *Proc. AAAI Conf. Artificial Intelligence*, vol. 35, no. 9, 2021, pp. 7657–7666. [Online]. Available: https://ojs.aaai.org/index.php/AAAI/article/view/16936
[11] R. Hasani, M. Lechner, A. Amini, L. Liebenwein, A. Ray, M. Tschaikowski, G. Teschl, and D. Rus, "Closed-form continuous-time neural networks," *Nature Machine Intelligence*, vol. 4, no. 11, pp. 992–1003, Nov. 2022.
[12] D. Hafner, T. Lillicrap, J. Ba, and M. Norouzi, "Dream to control: Learning behaviors by latent imagination," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2020. [Online]. Available: https://arxiv.org/abs/1912.01603
[13] M. Janner, J. Fu, M. Zhang, and S. Levine, "When to trust your model: Model-based policy optimization," in *Advances in Neural Information Processing Systems*, vol. 32, 2019. [Online]. Available: https://proceedings.neurips.cc/paper/2019/hash/5faf461eff3099671ad63c6f3f094f7f-Abstract.html
[14] D. A. Pomerleau, "ALVINN: An autonomous land vehicle in a neural network," in *Advances in Neural Information Processing Systems*, vol. 1. Morgan Kaufmann, 1988. [Online]. Available: https://proceedings.neurips.cc/paper/1988/hash/812b4ba287f5ee0bc9d43bbf5bbe87fb-Abstract.html
[15] S. Ross, G. J. Gordon, and J. A. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," in *Proc. 14th Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, ser. Proceedings of Machine Learning Research, vol. 15. PMLR, 2011, pp. 627–635. [Online]. Available: https://proceedings.mlr.press/v15/ross11a.html
[16] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learning Representations (ICLR)*, 2015. [Online]. Available: https://arxiv.org/abs/1412.6980

Appendix A
Hyperparameters and Implementation Details

Table III lists every hyperparameter used in the paper. All training runs are reproducible from `configs/` in the source repository; estimator evaluation grids are listed verbatim in `configs/estimators/` and `configs/closed_loop/`.

Appendix B
Forward-Model Rollout Metrics

Table IV reports horizon-wise prediction error for the three forward-model variants used in the paper. Multi-step rollout supervision substantially reduces medium- and long-horizon error while paying a small cost at $h=1$.

Appendix C
Oracle-Supervised Correction-Gain Predictor (Diagnostic Upper Bound)

The reliability-adaptive observer of Sec. III-D sets $K_f(t)$ from agent-available signals only. As a non-agent-available upper bound we also train an offline-oracle-supervised correction-gain predictor and evaluate it on the same grid; the single-cell SPSA result of C2 effectively matches this predictor on its training cell.

a) Predictor architecture.: A small MLP $g_\phi : \mathbb{R}^8 \rightarrow [0,1]$ predicts the scalar correction gain from a condition feature vector. The input

$$\xi = \left[\,\mathrm{onehot}(c)\,;\ \sigma\,;\ d/d_{\max}\right] \in \mathbb{R}^8, \tag{17}$$

concatenates a 3-way controller one-hot $\mathrm{onehot}(c)$, the 4-dim noise-sigma vector σ (`qpos`, `qvel`, `tip_pos`, `reach_err`), and a normalised delay $d/d_{\max}$ with $d_{\max}=36$ steps. The network is a $8 \rightarrow 32 \rightarrow 32 \rightarrow 1$ MLP with ReLU and a sigmoid output.

TABLE III
HYPERPARAMETERS USED ACROSS THE PAPER.

Component	Hyperparameter	Value
Forward model f_θ	state dim action dim hidden layers activation output parameters	83 34 2 × 256 ReLU, LayerNorm residual (Δx) 118,355
Forward training	optimiser learning rate weight decay batch size epochs val split	Adam [16] 1×10^{-3} 0 256 200 (early stop patience 20) every 5th source episode
Multi-step loss	rollout horizon H	1, 4, 8
Stage A g_ϕ	input dim hidden layers output parameters	8 = 3 ctrl + 4 σ + 1 delay/$d_{\max}$ 2 × 32 sigmoid scalar in $[0, 1]$ 1,409
Stage A training	optimiser learning rate weight decay epochs	Adam 1×10^{-3} 1×10^{-3} 500
Joint-PD controller	K_p, K_d action scale α muscle map	30, 3 5 $\mathrm{clip}(\alpha\,\mathrm{ReLU}(M\,u_{\mathrm{joint}}))$, $M \in \mathbb{R}^{34\times 20}$
IK solver	method max iter / tol damping λ^2	damped Jacobian pseudoinverse 200 / 0.01 m 0.1
BC policy	state dim variants hidden layers output demos / training	83 (full) / 80 / 77 2 × 256 sigmoid 34-dim 30 ep. × 600 steps; 100 epochs
Eval grid (open-loop)	noise levels delays (steps) K values	none, low, medium, high, vhigh, xhigh 0, 6, 18, 36 0.0, 0.25, 0.5, 0.75, 1.0
Eval grid (closed-loop)	noise levels (focused) delays (focused)	none, high, xhigh 0, 18

TABLE IV
FORWARD-MODEL ROLLOUT MSE AND TIP PREDICTION ERROR ON THE HELD-OUT VALIDATION EPISODES. LOWER IS BETTER. ALL THREE MODELS SHARE THE SAME ARCHITECTURE AND TRAINING SET (51,534 TRANSITIONS ACROSS 106 EPISODES); THEY DIFFER ONLY IN THE ROLLOUT HORIZON H.

	MSE			Tip err. (m)		
Model	h=1	h=10	h=50	h=1	h=10	h=50
H=1 (single-step)	0.0055	0.0124	0.0524	0.0076	0.0355	0.1381
H=4 multi-step	0.0053	0.0084	0.0164	0.0088	0.0247	0.0635
H=8 multi-step	0.0055	0.0082	0.0114	0.0098	0.0215	0.0483

b) Oracles.: For each of the 72 conditions in the stress grid (3 controllers × 6 noise levels × 4 delays) we sweep $K \in K_{\mathrm{grid}} = \{0, 0.25, 0.5, 0.75, 1.0\}$ and select the K minimising a chosen criterion. We define an *open-loop* oracle (estimation accuracy) and a *closed-loop* oracle (downstream task error):

$$K^\star_{\mathrm{ol}}(\sigma, d, c) = \arg\min_{K \in K_{\mathrm{grid}}} \mathbb{E}_t\big[\|\hat{x}_t(K) - x_t\|^2 \,\big|\, \sigma, d, c\big], \quad (18)$$

$$K^\star_{\mathrm{cl}}(\sigma, d, c) = \arg\min_{K \in K_{\mathrm{grid}}} \mathbb{E}\big[\mathrm{minTip}(K) \,\big|\, \sigma, d, c\big], \quad (19)$$

where minTip is the closed-loop minimum tip-to-target distance over the episode. Two predictor variants g^{ol}_ϕ and g^{cl}_ϕ are trained against $K^\star_{\mathrm{ol}}$ and $K^\star_{\mathrm{cl}}$ respectively.

c) Closed-loop deployment.: Under the H=8 forward model on the focused 6-cell joint-PD grid, g^{ol}_ϕ outputs $K \approx 0.36$–0.99 and trails K=0 by 14–28 cm (the open-loop oracle never selects K=0 because the open-loop estimation error is minimised by trusting the observation when the buffer rollout is long). The closed-loop sweep of Appendix E fixes this: $K^\star_{\mathrm{cl}} = 0$ at every cell, the retrained g^{cl}_ϕ outputs $K \approx 0.02$, and the predictor matches K=0 within 1.5 cm at the three delay-18 cells. A residual $\sim$ 17 cm gap persists at the three delay-0 cells because the closed-loop K-curve jumps sharply from K=0 to K=0.25 there (Appendix E) and the sigmoid asymptote 0.02 lies on the wrong side of the jump. Across the six cells, g^{cl}_ϕ closes 40–60% of the g^{ol}_ϕ-vs-K=0 gap with no architectural change to the predictor.

d) Transfer to the endpoint-error sensorimotor policy.: Repeating the closed-loop deployment with controller (iii) (endpoint-error sensorimotor policy, Sec. III, item iii) shows the same qualitative pattern, in a weaker form: K=0 beats K=1 by -18.5 to -18.7 cm at the three d=0 cells and the two are within ± 3.2 cm at d=18 (Table V). The endpoint-error policy's K=0 baseline is $\sim$11 cm worse in absolute terms than joint-PD's because the Jacobian-transpose projection is locally correct only; the qualitative pattern (closed-loop $K^\star = 0$ on this grid) is preserved.

TABLE V
ORACLE-SUPERVISED PREDICTOR DEPLOYED ON THE ENDPOINT-ERROR SENSORIMOTOR POLICY (III) ON THE FOCUSED 6-CELL GRID UNDER THE H=8 FORWARD MODEL. MEAN MIN-TIP (M) OVER 10 EPISODES PER CELL. THE "LEARNED-CL" COLUMN IS g^{cl}_ϕ TRAINED ON JOINT-PD $K^\star_{\mathrm{cl}}$ LABELS.

noise	delay	K=0	K=1	learned-cl ($\bar{K} \approx 0.02$)
none	0	0.604	0.789	0.754
none	18	0.647	0.631	0.687
high	0	0.604	0.789	0.693
high	18	0.647	0.614	0.652
xhigh	0	0.604	0.791	0.656
xhigh	18	0.647	0.628	0.628

e) Behaviour-cloned policy washes out the signal.: The behaviour-cloned controller (iv) (a two-hidden-layer 256-unit sigmoid MLP trained on 30 scripted IK demonstrations) reaches better than the joint-PD baseline in absolute terms ($\mathrm{success}_{<0.10\,\mathrm{m}}$ from $\sim$ 12% to $\sim$ 30% in the easier cells) but the gap between any two estimators collapses to under 3 cm. We take this as evidence that the BC policy at this demonstration scale is effectively open-loop in the state channel and report it as a methodological caution for closed-loop estimator benchmarks: the downstream controller must be demonstrably state-coupled, or any observer differences will not surface. The behaviour-cloned controller is therefore retained only as a diagnostic and is not used in the main C1–C4 results.

f) Why we keep this as an upper-bound baseline only.: g^{cl}_ϕ requires a per-condition K-sweep that the biological agent does not have access to. We report it as the strongest baseline that uses non-agent-available information; the C2 single-cell

SPSA result matches it on the training cell while using only innovation history and per-episode outcome, both of which the agent can plausibly compute on-line.

APPENDIX D
FULL OPEN-LOOP ORACLE $K^\star$ GRIDS

Table VI reports the mean oracle $K^\star$ per (delay, noise) cell under the H=8 multi-step-supervised forward model used throughout the main results, supervising the open-loop predictor variant g_ϕ^{ol} of Appendix C. Under single-step supervision (H=1) the $K^\star$<1 regime is confined to zero delay (18 of 72 cells); H=4 supervision widens the regime to 26 of 72 cells. The qualitative pattern (column-wise decrease at d=0, column-wise increase with delay) is the same across all three forward-model variants.

TABLE VI
ORACLE CORRECTION GAIN $K^\star$ AS A FUNCTION OF DELAY AND NOISE UNDER THE H=8 MULTI-STEP-SUPERVISED FORWARD MODEL (MEAN OVER 3 CONTROLLERS; 72 CONDITIONS TOTAL). 27 OF 72 CELLS REQUIRE $K^\star$<1.

	noise					
delay	none	low	medium	high	vhigh	xhigh
0	1.000	0.833	0.667	0.417	0.333	0.250
6	1.000	1.000	1.000	0.833	0.750	0.500
18	1.000	1.000	1.000	0.917	0.917	0.833
36	1.000	1.000	1.000	1.000	1.000	0.833

APPENDIX E
CLOSED-LOOP K-SWEEP ON THE FOCUSED GRID

Table VII reports the closed-loop $\min$-tip-to-target error per cell as a function of the (fixed) correction gain K, under the H=8 forward model and the joint-PD controller (10 episodes per cell). This K-sweep grounds two arguments in the main text: it defines the closed-loop oracle $K_{\mathrm{cl}}^\star$ used by the diagnostic predictor of Appendix C, and it is the baseline against which the reliability-adaptive observer is compared in Sec. IV-B–IV-C. $K_{\mathrm{cl}}^\star$=0 in every cell of the focused grid.

Two regularities are worth noting. First, at zero delay the K-curve is sharp around K=0: moving from K=0 to K=0.25 jumps $\min$-tip from 0.49 m to 0.77 m. The oracle-supervised predictor's sigmoid asymptote ($K \approx 0.025$, Appendix C) sits on the wrong side of this jump in delay-0 cells, which is why the diagnostic gap remains ($\sim$17 cm). Second, at delay-18 the K-curve is smooth near K=0 (K=0 and K=0.25 within 0.01–0.05 m) so the predictor's small offset is harmless and the same predictor matches the K=0 baseline within 1.5 cm. These regularities also explain why the reliability-adaptive observer of C1 (which floats around $K_f \approx 0.6$–0.7) does badly at every delay-18 cell and at all delay-0 cells alike: any non-trivial K lands in the high-error plateau.

APPENDIX F
TRANSFER TEST ON REACHRANDOM

Setup. We re-run a slim version of the pipeline on `myoArmReachRandom-v0` (same 34-muscle arm,

TABLE VII
CLOSED-LOOP $\min$-TIP ERROR (M) PER CELL ACROSS THE K-SWEEP (CORRECTION GAIN) UNDER THE H=8 FORWARD MODEL AND THE JOINT-PD CONTROLLER. EACH CELL AVERAGES 10 EPISODES. BOLD MARKS THE PER-CELL MINIMUM.

noise	delay	K=0	K=0.25	K=0.5	K=0.75	K=1
none	0	**0.493**	0.771	0.782	0.774	0.770
none	18	**0.497**	0.498	0.708	0.746	0.697
high	0	**0.493**	0.767	0.780	0.771	0.770
high	18	**0.497**	0.538	0.696	0.746	0.691
xhigh	0	**0.493**	0.772	0.769	0.766	0.762
xhigh	18	**0.497**	0.550	0.713	0.749	0.700

same 600-step episodes, random per-episode target instead of a fixed one). Targets come from the same `runs/targets/train.npz` file (200 positions) so that targets are paired across estimators and forward models; initial joint pose is deterministic (q=0) for both variants. Transition data: 49,740 transitions across 106 episodes (random / lowamp / hold controllers), comparable to the 51,534- transition ReachFixed dataset. The marginal C2 cell-level effect ($\approx$ 8.6 cm at one cell with learned $K \approx 0.99$ and paired-t p=0.21) is task-dependent and *not* treated as a transfer target; the test asks whether the structural phenomena C1 / C3 / C4 persist under variable-target reaching. Pre-committed pass criteria, fixed before reading the numbers:

- *C1 replicate*: open-loop oracle $K_{\mathrm{ol}}^\star$ varies with noise / delay, with long-delay cells biasing toward K=1.
- *C3 replicate*: multi-step supervision reduces h=50 forward-model error and increases the share of closed-loop cells where K=0 or prediction-heavy estimators beat K=1.
- *C4 replicate*: a closed-loop oracle $K_{\mathrm{cl}}^\star$ trained predictor moves toward the K=0 / $K_{\mathrm{cl}}^\star$ side of the K-curve compared with an open-loop oracle predictor.

Forward-model rollout error (Table VIII) mirrors the ReachFixed pattern: H=8 multi-step rollout supervision reduces h=50 tip prediction error by 48% versus H=1 (0.110 $\to$ 0.057 m).

TABLE VIII
REACHRANDOM FORWARD-MODEL ROLLOUT MSE AND TIP PREDICTION ERROR ON HELD-OUT VALIDATION EPISODES. SAME NETWORK AND TRAINING RECIPE AS THE MAIN TEXT; LOWER IS BETTER.

	MSE			Tip err. (m)		
Model	h=1	h=10	h=50	h=1	h=10	h=50
H=1 (single-step)	0.0045	0.0103	0.0318	0.0066	0.0291	0.1098
H=8 multi-step	0.0046	0.0072	0.0096	0.0093	0.0229	0.0573

Closed-loop reaching, single-step model. Table IX reports $\min$-tip error at the 6-cell focused grid for K=0 and K=1 under the joint-PD controller. Two observations: (i) K=0 is constant across noise/delay because it ignores observations, and (ii) on ReachRandom K=0 already beats K=1 in *every* cell, by 7–27 cm. This contrasts with ReachFixed where the H=1 closed-loop optimum was nearer K=1 at delay-18 cells; the gap is therefore task-dependent at the single-step baseline.

TABLE IX
REACHRANDOM H=1 CLOSED-LOOP min-TIP ERROR (M, MEAN OVER 10 EPISODES) UNDER THE JOINT-PD CONTROLLER. K=0 BEATS K=1 IN EVERY CELL BY 7–27 CM.

noise	delay	K=0	K=1	K=1−K=0
none	0	0.499	0.770	+27.2 cm
none	18	0.498	0.578	+8.0 cm
high	0	0.499	0.770	+27.2 cm
high	18	0.498	0.567	+6.8 cm
xhigh	0	0.499	0.762	+26.3 cm
xhigh	18	0.498	0.593	+9.4 cm

Closed-loop K-curve, H=8 multi-step model. Table X reports the full $K \in \{0, 0.25, 0.5, 0.75, 1\}$ sweep at H=8. The closed-loop oracle $K^\star_{\rm cl}$ is the per-cell minimiser. Unlike ReachFixed, where $K^\star_{\rm cl}$=0 in every cell of the focused grid, on ReachRandom $K^\star_{\rm cl}$ splits across three values: K=0 at all three d=0 cells (matching ReachFixed), K=0.25 at delay-18 in the low/medium-noise cells, and K=1 at the (xhigh, d=18) cell.

TABLE X
REACHRANDOM H=8 CLOSED-LOOP min-TIP K-SWEEP (MEAN OVER 10 EPISODES). BOLD MARKS THE PER-CELL $K^\star_{\rm cl}$.

noise	delay	K=0	K=0.25	K=0.5	K=0.75	K=1
none	0	**0.670**	0.762	0.785	0.780	0.770
none	18	0.642	**0.625**	0.706	0.644	0.647
high	0	**0.670**	0.764	0.785	0.775	0.770
high	18	0.642	**0.631**	0.688	0.651	0.639
xhigh	0	**0.670**	0.758	0.765	0.761	0.762
xhigh	18	0.642	0.662	0.691	0.650	**0.627**

Open-loop oracle structure on ReachRandom. We re-ran the full 3 controllers × 6 noise × 4 delay × 5 K stress eval (using 2 episodes per controller, matching the ReachFixed scale) for both forward models. $K^\star_{\rm ol} < 1$ holds in 18 / 72 cells under H=1 (exactly matching ReachFixed at H=1) and 25 / 72 cells under H=8 (vs. 27 / 72 on ReachFixed); blending is again confined to delay-0 cells under H=1 and expands to delay $\geq$ 6 under H=8 (Table XI). C1 therefore replicates structurally on ReachRandom.

Learned predictors trained on $K^\star_{\rm ol}$ vs $K^\star_{\rm cl}$. We train the same 8→32→32→1 Stage A predictor on (i) the $K^\star_{\rm ol}$ table from the H=8 ReachRandom stress eval, and (ii) the per-cell $K^\star_{\rm cl}$ labels from Table X (extended to the 72-cell schema by replicating the joint-PD label across the auxiliary controllers and the unswept delays/noises). Table XII reports the K each predictor outputs at the focused grid and the resulting closed-loop min-tip. The $K^\star_{\rm ol}$ predictor recovers values near K=1 at delay-18 cells (its training labels were $K^\star_{\rm ol}$=1 there), matching the K=1 baseline within 1 cm. The $K^\star_{\rm cl}$ predictor recovers small K (0.06–0.31) at delay-0 cells, beating the $K^\star_{\rm ol}$ predictor by 1.3–12.6 cm there. At delay-18 the $K^\star_{\rm cl}$ predictor's outputs drift toward intermediate values (K=0.24–0.95) and the resulting closed-loop reach is comparable to K=1 (within ±8 cm). The oracle-redefinition transfer therefore holds clearly at delay-0 but is muted at delay-18 because the closed-loop K-curve there is relatively flat (Table X) and the sigmoid output cannot cleanly land on the mixed $K^\star_{\rm cl}$ (0/0.25/1) values across the three noise rows.

TABLE XI
REACHRANDOM $K^\star_{\rm ol}$ (MEAN OVER 3 CONTROLLERS) PER (NOISE, DELAY) CELL, UNDER H=1 (TOP) AND H=8 (BOTTOM). DASHES MARK $K^\star_{\rm ol}$=1.0 (TRUST THE OBSERVATION). THE NON-TRIVIAL $K^\star_{\rm ol} < 1$ REGIME IS CONFINED TO DELAY-0 UNDER H=1 AND SPREADS TO DELAY ≥ 6 UNDER H=8, MIRRORING REACHFIXED.

H=1 single-step forward model				
noise \ delay	0	6	18	36
none	–	–	–	–
low	0.83	–	–	–
medium	0.67	–	–	–
high	0.50	–	–	–
vhigh	0.33	0.92	–	–
xhigh	0.25	0.67	–	–
H=8 multi-step forward model				
noise \ delay	0	6	18	36
none	–	–	–	–
low	0.83	–	–	–
medium	0.67	–	–	–
high	0.42	0.92	–	–
vhigh	0.33	0.67	0.92	–
xhigh	0.25	0.50	0.75	0.75

TABLE XII
REACHRANDOM H=8 CLOSED-LOOP COMPARISON OF FOUR ESTIMATORS ON THE FOCUSED GRID. COLUMNS: K=0, K=1, *learned-ol* (STAGE A TRAINED ON $K^\star_{\rm ol}$), *learned-cl* (STAGE A TRAINED ON $K^\star_{\rm cl}$); FOR EACH LEARNED PREDICTOR WE REPORT BOTH ITS MEAN OUTPUT $\bar{K}$ AND THE CLOSED-LOOP min-TIP (M, MEAN OVER 10 EPISODES).

noise	delay	K=0	K=1	learned-ol		learned-cl	
				$\bar{K}$	m	$\bar{K}$	m
none	0	0.670	0.770	0.80	0.781	0.06	0.655
none	18	0.642	0.647	0.98	0.638	0.24	0.673
high	0	0.670	0.770	0.71	0.777	0.09	0.690
high	18	0.642	0.639	0.96	0.644	0.47	0.721
xhigh	0	0.670	0.762	0.36	0.783	0.31	0.770
xhigh	18	0.642	0.627	0.88	0.654	0.95	0.642

Assessment against the pre-committed criteria.

- *C1 (full replication)*: open-loop oracle $K^\star_{\rm ol}$ varies with noise and delay in the same qualitative shape as ReachFixed, with the $K^\star_{\rm ol} < 1$ fraction growing from 18 / 72 (H=1) to 25 / 72 (H=8).
- *C3 (partial replication)*: H=8 rollout supervision reduces forward-model error by 48 % (replicates) and K=0 is optimal at all three delay-0 cells (replicates). The ReachFixed-style *regime shift* from K=1-winning at H=1 to K=0-winning at H=8 does *not* apply because K=0 was already winning at H=1; the variable-target variant is structurally more favourable to prediction-only across forward-model strengths.
- *C4 (partial replication, made richer)*: the closed-loop oracle $K^\star_{\rm cl}$ on ReachRandom is *not* uniformly K=0 (3 cells pick K=0, 2 cells pick K=0.25, 1 cell picks K=1), confirming $K^\star_{\rm cl}$ is a non-degenerate task-and-cell-dependent function. A Stage A predictor retrained on these labels matches the K=0 baseline within 1.3 cm at

every delay-0 cell and beats the $K^\star_{\text{ol}}$-trained predictor by 1–13 cm there. At delay-18, where $K^\star_{\text{cl}}$ takes three different values across the three noise rows, the predictor's sigmoid output cannot cleanly land on each label, and the closed-loop benefit is muted.

Take-away. On ReachRandom the closed-loop oracle $K^\star_{\text{cl}}$ is multimodal across cells ($\{0, 0.25, 1\}$), in contrast to the uniform $K^\star_{\text{cl}} = 0$ on ReachFixed. The diagnostic oracle-supervised predictor g^{cl}_ϕ of Appendix C transfers cleanly to delay-0 cells; at delay-18 the sigmoid-output parameterisation cannot fully exploit the cell-dependent $K^\star_{\text{cl}}$. The same heterogeneity drives the C4 finding of Sec. IV-E: *default within-trial reliability* (no SPSA training of β) beats $K{=}0$ at delay-18 cells on ReachRandom because its $K_f \approx 0.6$– 0.7 accidentally lands near the per-cell oracle values, while the ReachFixed-trained β pushes K_f globally lower and worsens performance on ReachRandom.

Appendix G
Software, Hardware, and Reproducibility

All experiments run on a single workstation (no GPU is required; the forward model and Stage A predictor are small enough that CPU training takes only minutes). Software stack: Python 3.11, PyTorch (CPU), MuJoCo 2.3, MyoSuite 2.12.2, Gymnasium 0.29. Reproduction scripts and exact dataset paths are recorded in the source repository's `docs/02_InitialImplementationPlan.md` and `docs/03_PaperOutline.md`.